

Citation Anomaly Detection

via Lorentzian Embeddings and Temporal Attention

Project Code
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Students
Nadir Yaakov, Ben Zacai

Supervisor
Dr. Dvora Toledano-Kitai

BACKGROUND & MOTIVATION

Scientific citation networks are growing exponentially, leading to citation manipulation patterns such as:

- Citation Cartels
- Coordinated Spikes
- Excessive Self-Citation

THE CHALLENGE

Traditional Euclidean embedding methods (e.g., standard GCNs) struggle to capture the hierarchical and tree-like structure of citation networks, resulting in high distortion and poor detection of structural anomalies.

THE SOLUTION: HYPERBOLIC GEOMETRY & LLGC

- Citation snapshots are embedded into the Lorentzian manifold** (hyperbolic space), naturally accommodating hierarchical relationships with minimal distortion.
- Temporal segmentation** into 5-year windows tracks embedding evolution.
- Structural Drift:** Comparing embeddings of each paper across time highlights deviations that signal anomalous behavior.

Note: Hyperbolic embeddings outperform Euclidean approaches for hierarchical networks.

SYSTEM ARCHITECTURE & METHODOLOGY

Data Preprocessing:

Semantic Scholar data integration, TF-IDF on abstracts, numerical standardization, Field-of-Study extraction.

Graph Construction:

Building dynamic graphs where nodes are papers and edges are citations.

Embedding Generation:

Training the LLGC model to learn node representations in hyperbolic space.

Anomaly Detection:

Utilizing the Isolation Forest classifier to detect outliers based on embedding shifts and structural features.

VALIDATION

Real Citation Data



LLGC Training



Synthetic Anomaly Injection

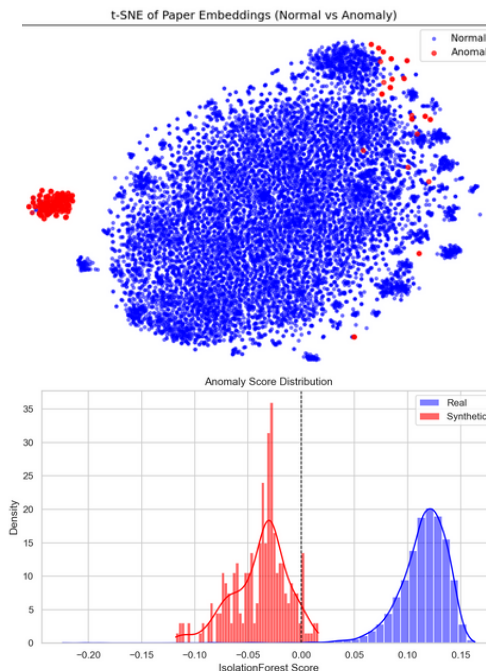


Unsupervised Detection



Quantitative Evaluation

VALIDATION EXPERIMENTAL RESULTS



VALIDATION EVALUATION RESULTS:

Precision: 0.83
F1-Score: 0.88
Recall: 0.94

CONCLUSIONS

- Hyperbolic geometry provides a superior framework for modeling citation networks and identifying structural anomalies.
- Temporal segmentation allows detection of evolving anomalies without retraining on synthetic data (Out-of-Distribution detection).
- The model provides actionable insights for publishers to maintain citation integrity and identify potential manipulative practices.

FUTURE DIRECTIONS

Integration of Temporal Attention Mechanisms to dynamically weight historical snapshots and improve detection of long-term citation anomalies.